Learning LaTeX Through a Simple Article

Kotapati Sai Mounika B24CS019

04/09/2025

1 Introduction

This article is designed to show how a complete LaTeX document looks. It demonstrates document classes, titles, sectioning, lists, and mathematical symbols.

1.1 Motivation

LaTeX is used for producing professional documents.

1.1.1 Why Students Need It

Because it ensures formatting is consistent.

Key Point Learning LaTeX early saves time in research.
Mini Note Even small reports benefit from LaTeX.

2 Organizing Information

Information is often presented as lists. LaTeX supports both itemized and numbered lists.

2.1 Custom Itemize

- Normal bullet

  Urgent task

■ Completed work

NOTE Special note

2.2 Numbered List

1. First numbered point

   Custom exclamation

■ Black square instead of number

NOTE Replace number with text

2.3 Nested Example

1. Outer Level

   - Inner bullet

Note: This is a note
Wait! This is a warning

3 Mathematics in Action

By practicing this, you now cover all the essential commands from the tutorial.

3.1 Summation Example

$$S_n = \frac{1}{n} \sum_{k=1}^{n} \frac{k^2 + k + 1}{k^2 + 1}$$

3.2 Geometric Series

$$1 + r + r^2 + r^3 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}$$